American Express Data Analyst Interview Questions (0-3 Years) 17-19 lpa

SQL Questions

1. Write a query to find duplicate rows in a table.

To detect duplicates, identify columns that should be unique and group by them.

Example:

```
SELECT column1, column2, COUNT(*) AS count
FROM your_table
GROUP BY column1, column2
HAVING COUNT(*) > 1;
```

Explanation:

- GROUP BY combines rows with the same values in the specified columns.
- HAVING COUNT(*) > 1 filters those combinations that occur more than once, indicating duplicates.

Tip: Add ROW_NUMBER() or RANK() with CTE to highlight or delete duplicates if needed.

2. Explain the difference between INNER JOIN and OUTER JOIN with examples.

INNER JOIN:

Returns only **matching** records from both tables.

```
SELECT e.name, d.department_name
FROM employees e
INNER JOIN departments d ON e.department_id = d.department_id;
```

- Output: Only employees who belong to a department.

LEFT OUTER JOIN:

Returns **all records from the left** table, and matching records from the right table. If no match, NULL is returned.

```
SELECT e.name, d.department_name
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id;
```

- Output: All employees, with department info where available.

RIGHT OUTER JOIN:

Returns **all records from the right** table, and matching records from the left.

FULL OUTER JOIN:

Returns **all records from both tables**, matching where possible.
Key Difference:

- INNER JOIN = intersection (matched data only)
- OUTER JOIN = union + NULLs (matched + unmatched data)

3. Write a query to fetch the second-highest salary from an employee table.

Option 1: Using DISTINCT, ORDER BY, and LIMIT (MySQL/PostgreSQL)

```
SELECT DISTINCT salary
FROM employees
ORDER BY salary DESC
LIMIT 1 OFFSET 1;
```

Option 2: Using subquery (Generic SQL)

```
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

Explanation:

- The subquery fetches the highest salary.
- The outer query finds the maximum salary **less than** the highest — giving the second-highest.

4. How do you use GROUP BY and HAVING together? Provide an example.

Use GROUP BY to group data and HAVING to filter **aggregated results** (unlike WHERE, which filters raw rows).

```
SELECT department_id, COUNT(*) AS emp_count
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5;
```

Explanation:

- Groups employees by department.
- Filters groups where the count of employees is **more than 5**.

5. Write a query to find employees earning more than their managers.

Assume the table employees has:
emp_id, name, salary, manager_id

```
SELECT e.name AS employee_name, e.salary, m.name AS manager_name, m.salary AS
manager_salary
FROM employees e
JOIN employees m ON e.manager_id = m.emp_id
WHERE e.salary > m.salary;
```

Explanation:

- Self-join: matches employees (e) with their managers (m).
- Filters those where employee's salary > manager's salary.

6. What is a window function in SQL? Provide examples of ROW_NUMBER and RANK.

Definition:

A **window function** performs calculations **across a set of table rows** related to the current row — without collapsing rows like GROUP BY.

Syntax:

FUNCTION_NAME() OVER (PARTITION BY column ORDER BY column)

Example: ROW_NUMBER()

Assigns a unique sequential number to each row **within a partition**.

```
SELECT name, department, salary,
    ROW_NUMBER() OVER (PARTITION BY department ORDER BY salary DESC) AS row_num
FROM employees;
```

- Each employee within the same department gets a row number based on salary rank (highest first).

Example: RANK()

Assigns **the same rank** to rows with **equal values**, but skips the next rank(s).

```
SELECT name, department, salary,
    RANK() OVER (PARTITION BY department ORDER BY salary DESC) AS rank_num
FROM employees;
```

- If 2 employees have the same salary, both get rank 1, and the next gets rank 3.

7. Write a query to fetch the top 3 performing products based on sales.

Assume table sales_data has:
product_id, product_name, total_sales

```
SELECT product_id, product_name, total_sales
FROM sales_data
ORDER BY total_sales DESC
LIMIT 3;
```

Alternate using RANK() (if ties matter):

```
SELECT product_id, product_name, total_sales
FROM (
  SELECT *, RANK() OVER (ORDER BY total_sales DESC) AS rank_num
  FROM sales_data
) ranked_sales
WHERE rank_num <= 3;
```

8. Explain the difference between UNION and UNION ALL.

Feature	UNION	UNION ALL
Duplicates	Removes duplicates	Keeps all rows, including duplicates
Performance	Slower (because of sorting)	Faster (no de-duplication)
Use case	When you want distinct rows	When duplicates are meaningful

Example:

SELECT city FROM customers
UNION
SELECT city FROM vendors;
→ Returns a unique list of cities.
SELECT city FROM customers
UNION ALL
SELECT city FROM vendors;
→ Returns **all cities**, including duplicates.

9. How do you use a CASE statement in SQL? Provide an example.

CASE lets you write conditional logic in SQL (similar to IF/ELSE).

```
SELECT name, salary,
      CASE
        WHEN salary >= 100000 THEN 'High'
        WHEN salary >= 50000 THEN 'Medium'
        ELSE 'Low'
      END AS salary_category
FROM employees;
```

Explanation:

- Assigns a category based on salary value.
- Works inside SELECT, WHERE, ORDER BY, etc.

10. Write a query to calculate the cumulative sum of sales.

Assume table sales has:
order_date, product_id, sales_amount

```
SELECT order_date, product_id, sales_amount,
      SUM(sales_amount) OVER (PARTITION BY product_id ORDER BY order_date) AS
cumulative_sales
FROM sales;
```

Explanation:

- SUM(...) OVER (...) calculates a **running total** per product based on order date.
- PARTITION BY groups by product, and ORDER BY ensures the accumulation follows chronological order.

11. What is a CTE (Common Table Expression), and how is it used?

Definition:

A **CTE (Common Table Expression)** is a temporary, named result set that you can reference within a SQL query.
It improves readability and simplifies complex subqueries or recursive logic.

Syntax:

```
WITH cte_name AS (
  SELECT ...
)
SELECT * FROM cte_name;
```

Example – Filter top-paid employees using CTE:

```
WITH HighEarners AS (
  SELECT emp_id, name, salary
  FROM employees
  WHERE salary > 100000
)
SELECT * FROM HighEarners;
```

Benefits:

- Reusable and readable
- Allows recursion (e.g., hierarchical data)
- Avoids repeating subqueries

12. Write a query to identify customers who have made transactions above $5,000 multiple times.

Assume transactions table has:
customer_id, transaction_amount

```
SELECT customer_id, COUNT(*) AS high_value_txns
FROM transactions
WHERE transaction_amount > 5000
GROUP BY customer_id
HAVING COUNT(*) > 1;
```

Explanation:

- Filters high-value transactions (> $5000).
- Groups them by customer.
- Returns customers who've done this **more than once**.

13. Explain the difference between DELETE and TRUNCATE commands.

Feature	DELETE	TRUNCATE
Removes rows	Yes (can use WHERE condition)	Yes (removes all rows)
WHERE supported?	Yes	No
Logging	Logs each deleted row (slower)	Minimal logging (faster)
Rollback	Can be rolled back (if within transaction)	Can be rolled back (in some RDBMS)
Identity reset	Retains identity	Resets identity (in most DBs)
Use case	Partial deletion or audit trail needed	Full data wipe without audit needed

14. How do you optimize SQL queries for better performance?

Here are **key SQL optimization techniques**:

1. Use SELECT only required columns

```
-- Bad
SELECT * FROM orders;
-- Good
SELECT order_id, customer_id FROM orders;
```

2. Create proper indexes

- Index frequently used columns in JOIN, WHERE, ORDER BY.

3. Avoid functions on indexed columns

```
-- Slower (cannot use index)
WHERE YEAR(order_date) = 2024
-- Better
WHERE order_date BETWEEN '2024-01-01' AND '2024-12-31'
```

4. Use EXISTS instead of IN (for subqueries)

```
-- Prefer EXISTS (better for large datasets)
SELECT name FROM customers c
WHERE EXISTS (
  SELECT 1 FROM orders o WHERE o.customer_id = c.customer_id
);
```

5. Avoid unnecessary joins or nested subqueries

6. Use appropriate data types and avoid implicit conversions

7. Analyze execution plans (EXPLAIN or EXPLAIN ANALYZE)

15. Write a query to find all customers who have not made

any purchases in the last 6 months.

Assume:

- customers(customer_id, name)
- transactions(customer_id, transaction_date)

```
SELECT c.customer_id, c.name
FROM customers c
LEFT JOIN transactions t
  ON c.customer_id = t.customer_id
    AND t.transaction_date >= CURRENT_DATE - INTERVAL '6 months'
WHERE t.customer_id IS NULL;
```

Explanation:

- LEFT JOIN includes all customers.
- WHERE t.customer_id IS NULL ensures the customer had **no purchase in the last 6 months**.

16. How do you handle NULL values in SQL? Provide examples.

NULL represents missing or unknown data.

1. Using IS NULL / IS NOT NULL:

```
SELECT * FROM employees WHERE manager_id IS NULL;
```

2. Replace NULL using COALESCE() or IFNULL() (MySQL):

```
SELECT name, COALESCE(phone_number, 'Not Provided') AS contact
FROM customers;
```

3. Handling NULLs in aggregation (e.g., AVG, SUM):

- These functions **ignore NULLs by default**.

```
SELECT AVG(salary) FROM employees;
```

4. Conditional checks:

```
SELECT name,
       CASE
         WHEN salary IS NULL THEN 'Unknown'
         ELSE 'Known'
       END AS salary_status
FROM employees;
```

17. Write a query to transpose rows into columns.

Assume a table sales with:
region, month, sales_amount
We want to **pivot month values** into columns.

Using CASE:

```
SELECT region,
       SUM(CASE WHEN month = 'Jan' THEN sales_amount ELSE 0 END) AS Jan,
       SUM(CASE WHEN month = 'Feb' THEN sales_amount ELSE 0 END) AS Feb,
       SUM(CASE WHEN month = 'Mar' THEN sales_amount ELSE 0 END) AS Mar
FROM sales
GROUP BY region;
```

Using PIVOT (SQL Server or Oracle syntax):

```
SELECT region, [Jan], [Feb], [Mar]
FROM (
  SELECT region, month, sales_amount
  FROM sales
) AS src
PIVOT (
  SUM(sales_amount)
  FOR month IN ([Jan], [Feb], [Mar])
) AS p;
```

18. Explain indexing and how it improves query performance.

What is an index?

An **index** is a data structure that improves the speed of data retrieval operations on a database table at the cost of additional space and write-time performance.

How indexing helps:

Feature	With Index	Without Index
Search performance	Fast (uses binary/tree search)	Slow (scans every row — full scan)
Used in	WHERE, JOIN, ORDER BY, GROUP BY	Inefficient for large datasets
Types	B-tree (default), Bitmap, Hash, etc.	-

Example:

```
-- Creating index
CREATE INDEX idx_customer_id ON transactions(customer_id);
```

- This helps queries like:

```
SELECT * FROM transactions WHERE customer_id = 101;
```

Important notes:

- Too many indexes can slow down INSERT/UPDATE.
- Avoid indexing columns with **low cardinality** (e.g., gender).
- Use **composite indexes** when querying multiple columns together.

19. Write a query to fetch the maximum transaction amount for each customer.

Assume a transactions table:

Column	Description
customer_id	ID of the customer
transaction_id	Unique transaction ID
amount	Transaction amount

Query:

```
SELECT customer_id, MAX(amount) AS max_transaction
FROM transactions
GROUP BY customer_id;
```

Explanation:

- GROUP BY groups all transactions by customer.
- MAX(amount) returns the highest transaction for each group (customer).

20. What is a self-join, and how is it used?

Definition:

A **self-join** is a regular join where a table is joined with itself.
It is useful when rows in a table are related to other rows in the same table.

Example Use Case – Employees and Managers:

Assume:

emp_id	name	manager_id
1	Alice	
2	Bob	1
3	Carol	1
4	David	2

Here, manager_id refers to emp_id of another employee.

Query: Get employee names along with their manager names

```
SELECT e.name AS employee_name, m.name AS manager_name
FROM employees e
LEFT JOIN employees m
  ON e.manager_id = m.emp_id;
```

Explanation:

- e is an alias for employees (as employee).
- m is another alias for the same table (as manager).
- The join links an employee to their manager using manager_id = emp_id.

Data Analysis/Scenario-Based Questions

21. How would you design a database to store credit card

transaction data?

To store credit card transaction data, we need to **normalize** the structure while keeping it **scalable**, **secure**, and **query-efficient**.

Suggested Schema Design:

1. Customers Table customer_id (PK), name, email, phone, address **2. Cards Table** card_id (PK), customer_id (FK), card_number (masked), card_type, status, issued_date **3. Merchants Table** merchant_id (PK), name, category, location **4. Transactions Table** transaction_id (PK), card_id (FK), merchant_id (FK), transaction_date, amount, currency, status, location

Best Practices:

- Mask sensitive fields (like card numbers).
- Store card_number as encrypted or tokenized.
- Use partitioning on date fields for faster querying.
- Add indexes on card_id, merchant_id, transaction_date.

22. Write a query to identify the most profitable regions based on transaction data.

Assume a transactions table:
(transaction_id, customer_id, amount, region, transaction_date)

Query to find top 3 profitable regions:

```
SELECT region, SUM(amount) AS total_revenue
FROM transactions
GROUP BY region
ORDER BY total_revenue DESC
LIMIT 3;
```

Explanation:

- Aggregates transaction amounts per region.
- Orders regions by total revenue.
- Retrieves top 3 using LIMIT.

Optional: You could also calculate profit by subtracting costs (if a cost column is present).

23. How would you analyze customer churn using SQL?

Step-by-step SQL approach:

Step 1: Define churn

Let's say a churned customer is one who hasn't transacted in the **last 6 months**.

Step 2: Sample schema

- customers(customer_id, name, signup_date)
- transactions(customer_id, transaction_date, amount)

Step 3: Query to identify churned customers

```
SELECT c.customer_id, c.name
FROM customers c
LEFT JOIN transactions t
  ON c.customer_id = t.customer_id
    AND t.transaction_date >= CURRENT_DATE - INTERVAL '6 months'
WHERE t.transaction_id IS NULL;
```

Step 4: Analyze churn metrics

You could extend this analysis by calculating:

- Churn rate = (Churned Customers / Total Customers) * 100
- Monthly churn trend
- Compare churned vs. active customers in terms of average spend

24. Explain the difference between OLAP and OLTP databases.

Feature	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing)
Purpose	Handles real-time transactional queries	Used for analytical/reporting queries
Operations	INSERT, UPDATE, DELETE	SELECT (aggregate, group, slice, dice)
Data Structure	Highly normalized (3NF)	De-normalized (star/snowflake schema)
Speed	Fast for read/write of single rows	Fast for complex analytical queries
Examples	Banking systems, e-commerce order processing	Business intelligence, dashboards, sales trends
Users	Clerks, DBAs	Analysts, Data Scientists
Backup/Recovery	Essential and frequent	Less frequent

In short:

- **OLTP** = operational, fast, real-time transactions.
- **OLAP** = analytical, slow-changing, historical data.

25. How would you determine the Average Revenue Per User (ARPU) from transaction data?

ARPU = Total Revenue / Total Number of Users

Assume a transactions table:

(transaction_id, customer_id, amount, transaction_date)

SQL Query:

```
SELECT
  SUM(amount) * 1.0 / COUNT(DISTINCT customer_id) AS ARPU
```

```
FROM transactions;
```

Explanation:

- SUM(amount) gets total revenue.
- COUNT(DISTINCT customer_id) counts unique users.
- Multiply by 1.0 to ensure float division.

You can also compute monthly ARPU by grouping by month.

```
SELECT
  DATE_TRUNC('month', transaction_date) AS month,
  SUM(amount) * 1.0 / COUNT(DISTINCT customer_id) AS monthly_arpu
FROM transactions
GROUP BY month
ORDER BY month;
```

26. Describe a scenario where you would use a LEFT JOIN instead of an INNER JOIN.

Use LEFT JOIN when:

You want **all records from the left table**, even if there's **no matching record** in the right table.

Real-life Scenario:

Question: List all customers and their transactions — even if they haven't made any.

Query:

```
SELECT c.customer_id, c.name, t.transaction_id, t.amount
FROM customers c
LEFT JOIN transactions t
  ON c.customer_id = t.customer_id;
```

Why LEFT JOIN?

- Shows **all customers**, including those with **no transactions** (returns NULLs for those).
- Using INNER JOIN would exclude customers with zero activity.

27. Write a query to calculate YoY (Year-over-Year) growth for a set of transactions.

Assume a table named transactions with:
(customer_id, transaction_date, amount)

Step 1: Extract year-wise revenue

```
SELECT
  EXTRACT(YEAR FROM transaction_date) AS year,
  SUM(amount) AS total_revenue
FROM transactions
GROUP BY EXTRACT(YEAR FROM transaction_date);
```

Step 2: Calculate YoY Growth using a CTE and Self-Join

```
WITH yearly_revenue AS (
```

```
 SELECT
    EXTRACT(YEAR FROM transaction_date) AS year,
    SUM(amount) AS total_revenue
  FROM transactions
  GROUP BY EXTRACT(YEAR FROM transaction_date)
)
SELECT
  curr.year AS current_year,
  curr.total_revenue,
  prev.total_revenue AS previous_year_revenue,
  ROUND(((curr.total_revenue - prev.total_revenue) / prev.total_revenue) * 100, 2) AS
yoy_growth_percent
FROM yearly_revenue curr
LEFT JOIN yearly_revenue prev
  ON curr.year = prev.year + 1;
```

Explanation:

- Joins each year to its previous year.
- Computes YoY growth as a percentage.

28. How would you implement fraud detection using transactional data?

Fraud detection typically involves pattern recognition, anomaly detection, and rule-based filtering.

Possible SQL-Based Checks:

Type	Rule
Unusual Amounts	Flag transactions > 3x average amount of that user
Rapid Repeats	Detect multiple transactions from same user within seconds
Location Mismatch	Transactions from different countries within a short time
Card Sharing	Same card used by different customers or IPs

Example Query – Unusual high amount per user:

```
WITH avg_txn AS (
  SELECT customer_id, AVG(amount) AS avg_amount
  FROM transactions
  GROUP BY customer_id
)
SELECT t.*
FROM transactions t
JOIN avg_txn a
  ON t.customer_id = a.customer_id
WHERE t.amount > 3 * a.avg_amount;
```

29. Write a query to find customers who have used more than 2 credit cards for transactions in a given month.

Assume a transactions table:
(customer_id, card_id, transaction_date)

Query:

```
SELECT customer_id,
       TO_CHAR(transaction_date, 'YYYY-MM') AS txn_month,
       COUNT(DISTINCT card_id) AS cards_used
FROM transactions
GROUP BY customer_id, TO_CHAR(transaction_date, 'YYYY-MM')
HAVING COUNT(DISTINCT card_id) > 2;
```

Explanation:

- Groups by customer_id and month.
- Counts distinct card_id used.
- Filters where more than 2 cards were used in a month.

30. How would you approach a business problem where you need to analyze the spending patterns of premium customers?

Step-by-Step Structured Approach:

Step 1: Understand the Objective

- Clarify with stakeholders what **"spending pattern"** means.
  - Is it frequency, amount, category, channel, or timing?
- Define **"premium customer"**:
  - Based on credit score, card tier (e.g., Platinum, Centurion), monthly spend threshold, etc.

Step 2: Data Collection

- Gather relevant datasets:
  - Customer table (ID, tier, demographics)
  - Transactions table (amount, date, category, location)
  - Cards table (card_type, limits, activation)

Step 3: Data Cleaning & Preparation

- Handle missing values and outliers.
- Filter only **premium customers** using defined criteria.
- Enrich data (e.g., categorize merchant types or locations).

Step 4: Exploratory Data Analysis (EDA)

Use SQL/Python/Power BI to derive insights like:

Focus Area	Example Analysis
Spend Amount	Average monthly/yearly spend
Time Trends	Seasonality or weekly spending behavior
Categories	Where they spend most (Travel, Dining, Shopping)
Geography	City or region-wise behavior

Focus Area	Example Analysis
Trends	Is their spend increasing/decreasing YoY?

Step 5: Segmentation

- Use clustering or thresholds to group premium customers into:
  - High spenders
  - Frequent spenders
  - Category loyalists (e.g., only travel)
- Identify anomalies or subgroups with unique patterns.

Step 6: Business Recommendations

- Personalize rewards or offers based on their dominant categories.
- Enhance retention strategies for segments showing decline.
- Promote premium card upgrades based on usage patterns.

Bonus: Sample SQL Query

Get top 3 spending categories of premium customers monthly:

```
SELECT customer_id,
       DATE_TRUNC('month', transaction_date) AS txn_month,
       category,
       SUM(amount) AS total_spend
FROM transactions
WHERE customer_id IN (
    SELECT customer_id FROM customers WHERE tier = 'Premium'
)
GROUP BY customer_id, txn_month, category
ORDER BY customer_id, txn_month, total_spend DESC;
```